FIGURE 1 — HEATMAP:

There is a distinct difference in severity by industry represented in this heatmap. Industries that belong to the Web, Online Advertising, Technology, Government and Finance sectors have a significantly higher number of records exposed relative to other industries.

This reinforces that Organization Type is an important factor in predicting breach severity and therefore was included in the regression modelling.

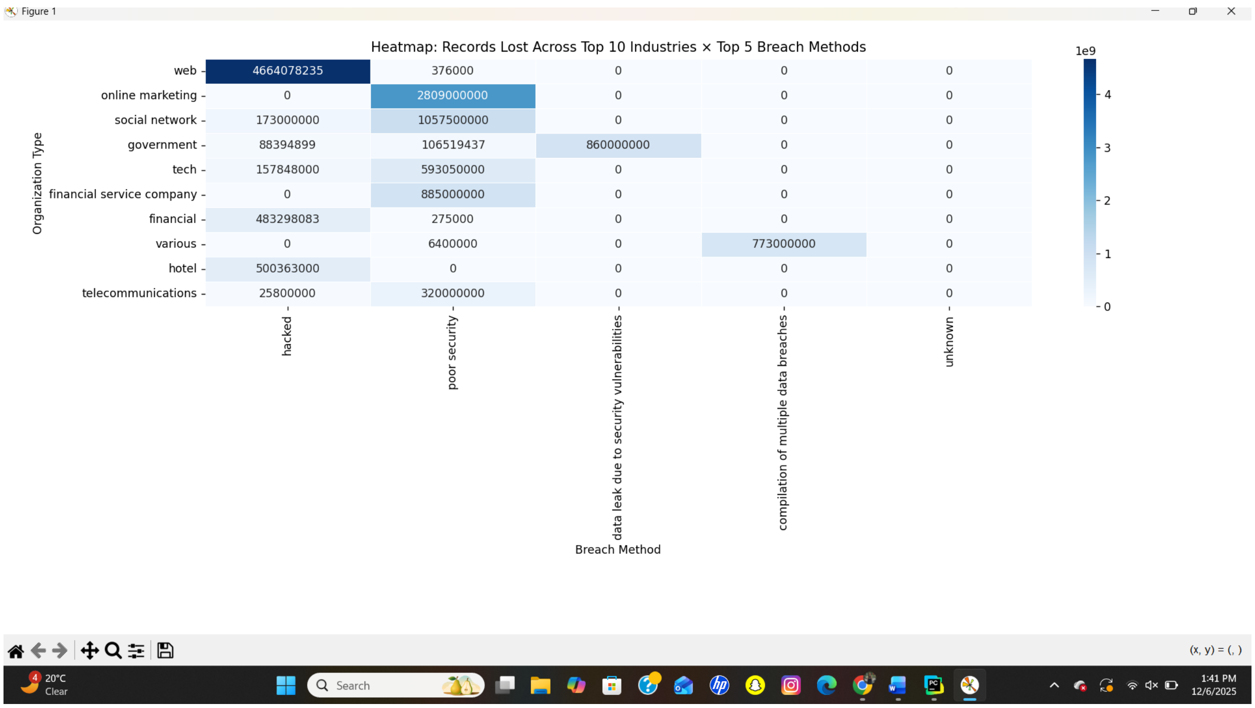


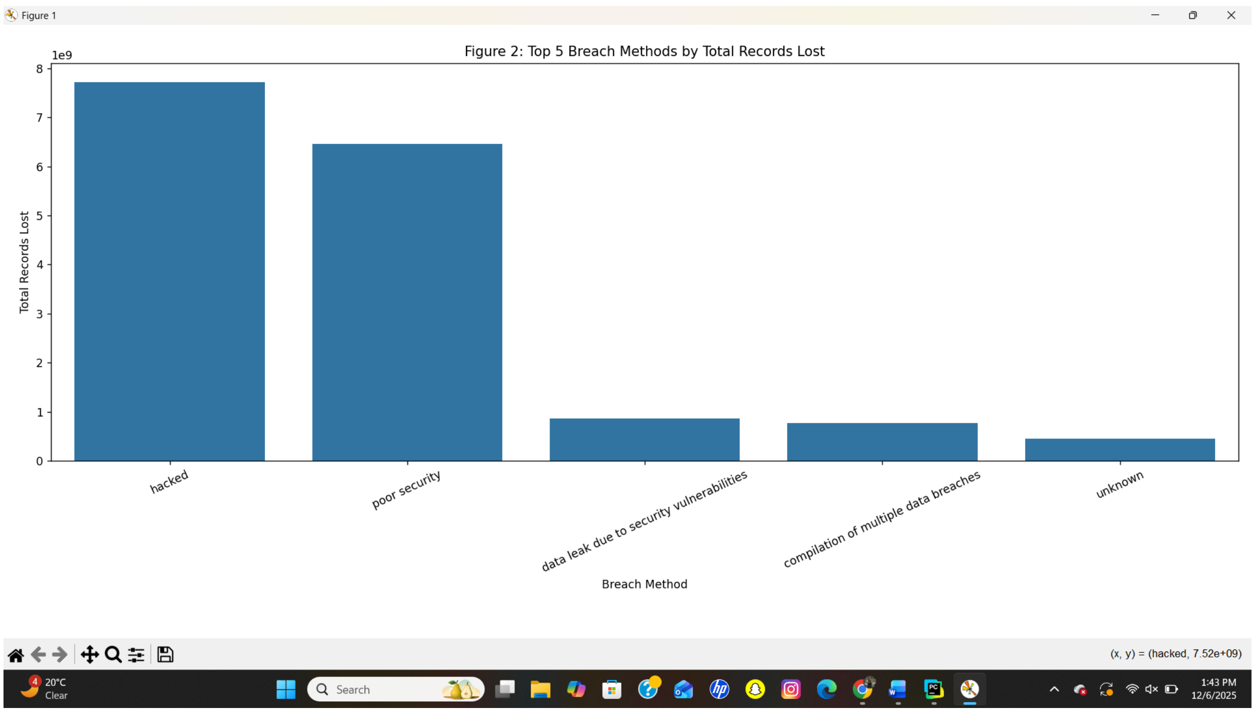
FIGURE 2 — TOP METHODS BAR CHART :

The bar chart also provides support for moving away from a Year-only analysis.

Hacking and Poor Security are the two methods responsible for the majority of breaches globally.

This shows that there are patterns of vulnerability within certain industries and through specific methods.

Therefore, industry-based variables have additional value beyond looking solely at time trends.



3.4 Analytical Approach:

There were two statistical models formulated. The first was a Simple Linear Regression based on the Year and number of Records. Records are equal to $\beta_0 + \beta_1$(Year)

The metrics based on the results of the OLS output are:

Figure 3 – Time Series Plot Figure 3 shows the total records lost per year and shows obvious spikes in certain years (2013, 2019).

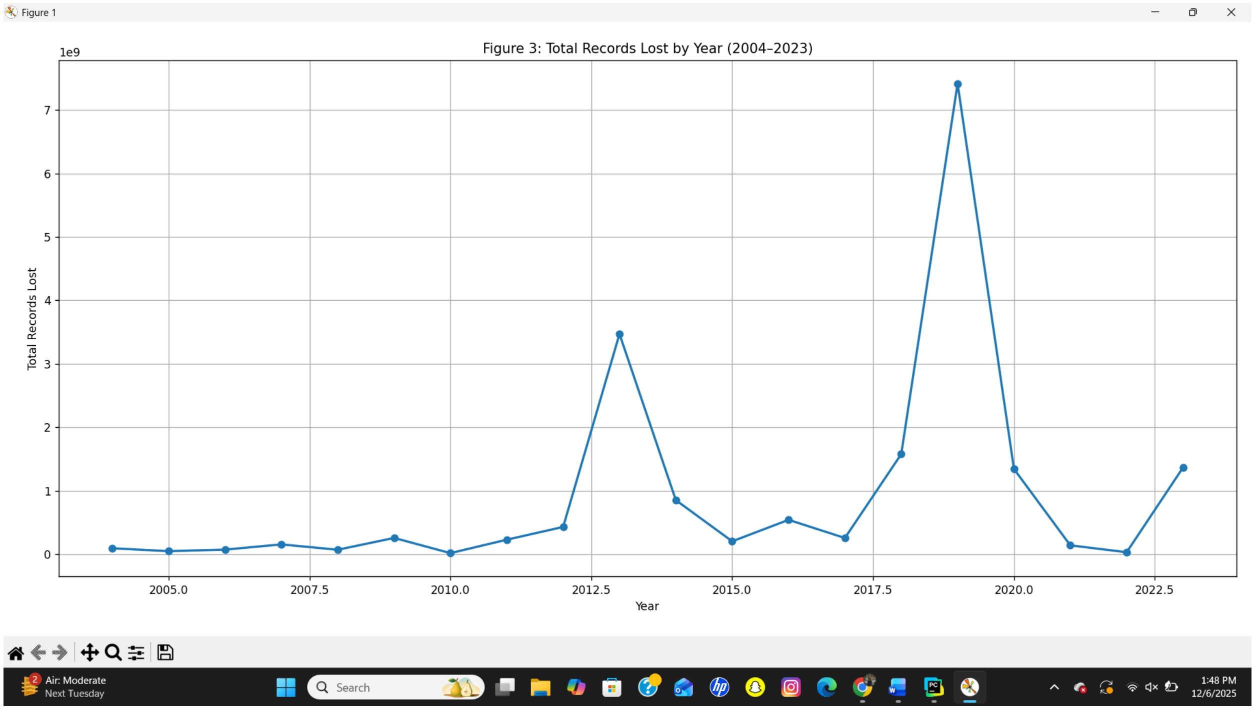
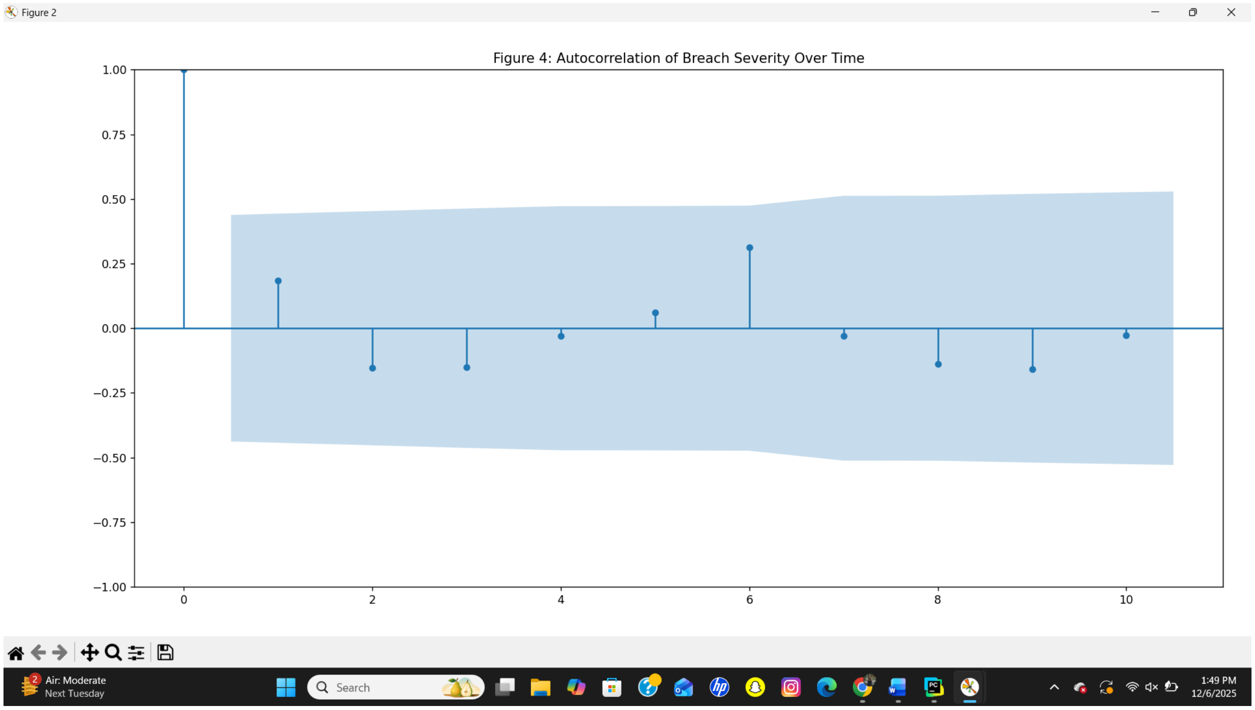


Figure 4 – Autocorrelation Plot Figure 4 provides information on no strong temporal correlation within the data .



Highlighted metrics from the Regression Output:

Metrics Values R^2 Approximately 0.011

P-value- Year approximately 0.039 (<0.05)

Slope Approximately +4.6 million Records per year

Conclusion: Severity demonstrates an evident increasing trend, yet the data also has a great deal of variance.

Model Two - Multiple Regression (Year + Industry = Records)

The industry categories were encoded numerically so that the regression could quantify the influence of each of the sectors.

Records = $\beta_0 + \beta_1$(Year) + β_2(Organization Type)

Metric Value R^2 ~ 0.0166 Year

p-value significant (<0.05)

Org_Type p-value weak/moderately (>0.05)*

Conclusion: The industry contributes some explanatory value, but its effect is still overshadowed by the more extreme outliers, indicating that breach severity is largely driven by events.

FIGURE 5 — SCATTER + TREND LINE

Regression trend confirms severity increases over time.

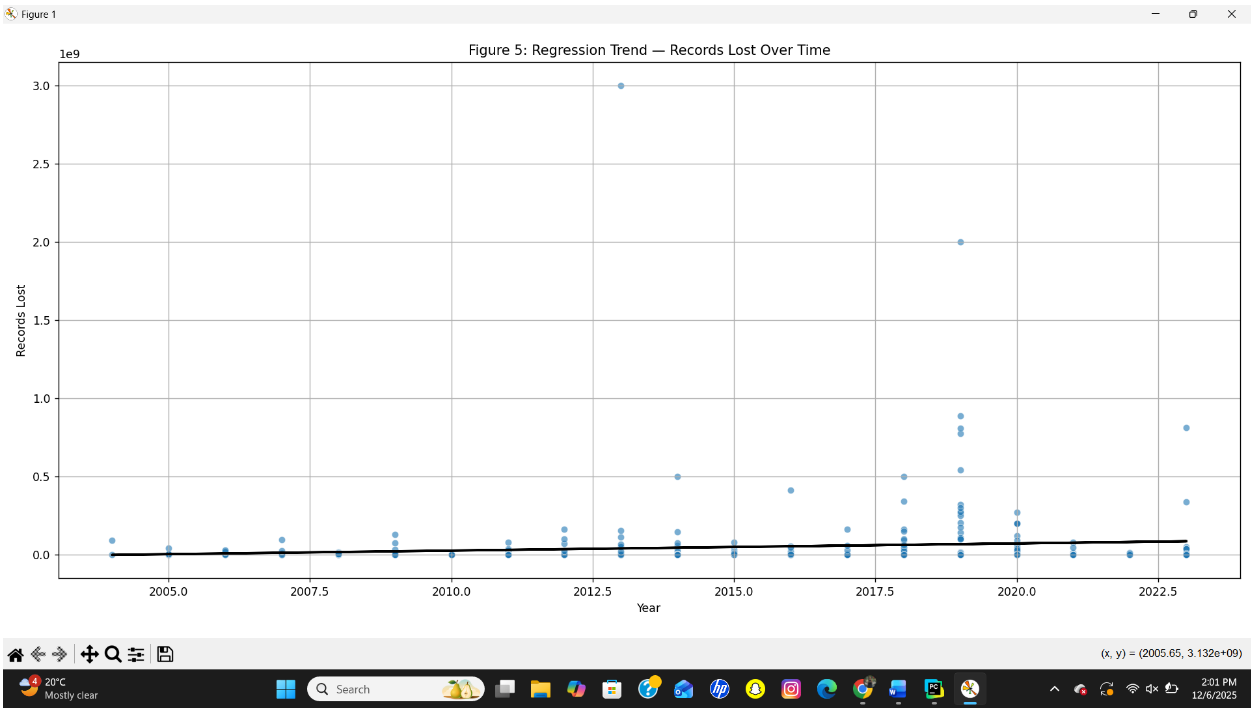


FIGURE 6 — FINAL HEATMAP

Sector × method interaction showing high-impact breach zones.

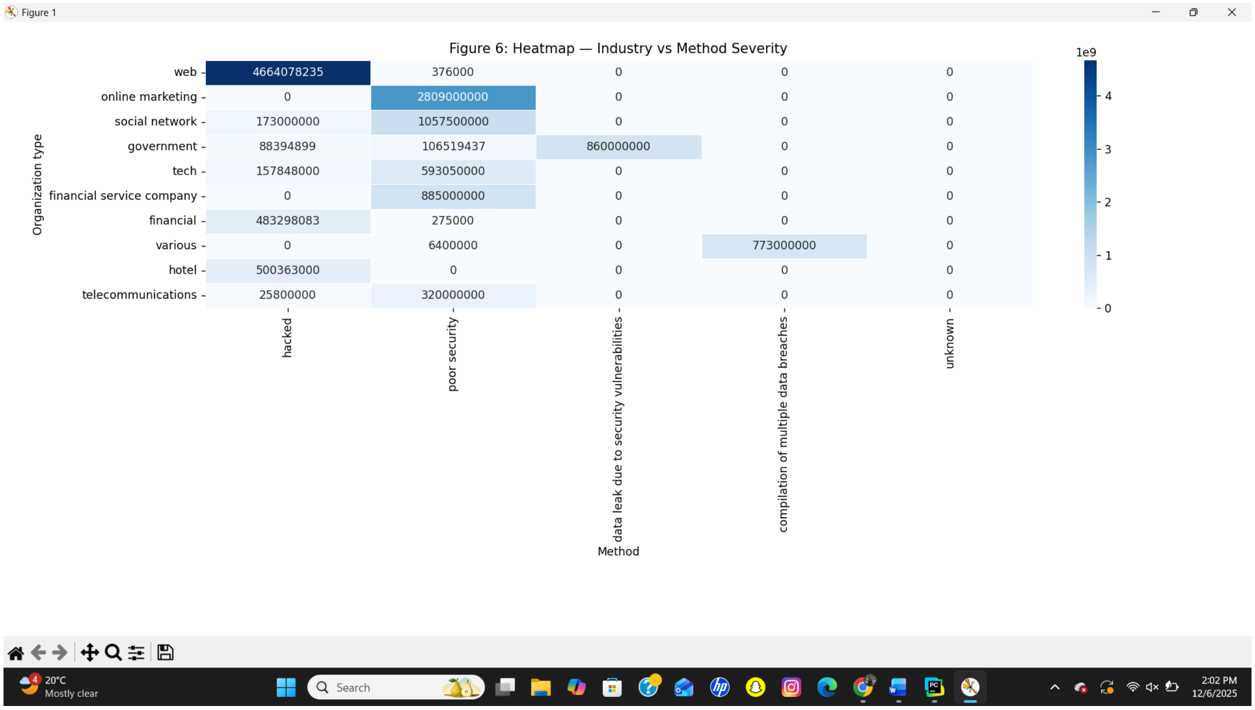


FIGURE 7 — TOP 5 METHODS BAR CHART

Hacking & poor security are leading causes of record exposure.